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Identifying infants at risk of sudden unexpected death with an automated predictive risk model

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ABSTRACT

Background/objective: Sudden unexpected infant death (SUID) is a common cause of infant death. We evaluated whether a predictive risk model (PRM) - Hello Baby - which was developed to stratify children by risk of entry into foster care could also identify infants at highest risk of SUID and non-fatal unsafe sleep events.
Participants and setting: Cases: Infants with SUID or an unsafe sleep event over 5½ years in a single county. Controls: All births in the same county.
Methods: Retrospective case-control study. Demographic and clinical data were collected and a Hello Baby PRM score was assigned. Descriptive statistics and the predictive value of a PRM score of 20 were calculated.
Results: Infants with SUID ($n = 62$) or an unsafe sleep event ($n = 37$) (cases) were compared with 23,366 births (controls). Cases and controls were similar for all demographic and clinical data except that infants with unsafe sleep events were older. Median PRM score for cases was higher than controls (17.5 vs. 10, $p < 0.001$); 50 % of cases had a PRM score 17–20 vs. 16 % of controls ($p < 0.001$).
Conclusions: The Hello Baby PRM can identify newborns at high risk of SUID and non-fatal unsafe sleep events. The ability to identify high-risk newborns prior to a negative outcome allows for individualized evaluation of high-risk families for modifiable risk factors which are potentially amenable to intervention. This approach is limited by the fact that not all counties can calculate a PRM or similar score automatically.

1. Introduction

Sudden unexpected infant death (SUID) is a common cause of death for US infants (*Linked birth/infant death records on CDC WONDER online database*, n.d.). After a major decrease in the incidence of sleep-related deaths immediately after the American Academy of Pediatrics released its 1992 recommendation that infants be placed on their back to sleep, the incidence has plateaued (Moon et al., 2022). Recognized risk factors for sleep-related deaths including prone sleeping and exposure to illicit substances have been incorporated into newborn education (Moon et al., 2022). Identification of specific infants at increased risk for SUID - which includes sudden infant death syndrome (SIDS), unexplained death and accidental suffocation and strangulation in bed (ASSB) - has

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Abbreviations

PRM	Predictive risk modeling
SUID	Sudden unexpected infant death

been difficult. As a result, universal education of parents and communities about unsafe sleep has been the main prevention strategy (Bechtel et al., 2020; Goodstein et al., 2015; Kellams et al., 2017; Moon et al., 2017). The ability to identify individual infants at highest risk for SUID would allow for targeted, tailored interventions.

Identification of infants at high-risk of non-fatal unsafe sleep events would allow for targeted and tailored intervention for this group as well. Unsafe sleep events can result in severe hypoxia or in contact injuries such as skull fracture. Bed and couch falls are recognized as an important cause of infant injury (Mulligan et al., 2017; Wegmann et al., 2016). While there are no data, to our knowledge, about how often bed falls occur while infants are asleep (vs. awake), one would hypothesize that the number is substantial given how much infants sleep and the prevalence of unsafe sleep. We are not aware of any data about the risk factors in infants who sustain injuries from non-fatal unsafe sleep events.

In Allegheny County Pennsylvania, a predictive risk model (PRM) tool was built to triage services and identify infants at risk of foster care placement due to maltreatment; the higher the score, the higher the risk of foster care placement by age 3 (Vaithianathan et al., 2019; Vaithianathan, Benavides-Prado, & Putnam-Hornstein, 2020). Hello Baby PRM risk score which ranges from 1 to 20. The score is calibrated so that ~5 % of a birth group falls within each score group. The PRM is already being used by Allegheny County to identify families at high risk of foster care placement due to maltreatment and proactively offer them a range of intensive services which address modifiable risk factors which can impact the likelihood of foster care placement. These services include home visitation, childcare, and housing support. The Hello Baby program is focused on strengthening families and trying to address the social determinants of health which lead to poor outcomes. It is not a punitive program and Hello Baby services are not mandated.

2. Methods

2.1. Subjects

2.1.1. Cases

SUID cases were identified by the county Child Death Review team which reviews all infant deaths. A death scene investigation and full autopsy is performed for every infant death in the county. We used the cause of death - R95 (SIDS), R99 (other ill-defied and unspecific causes of mortality) and W75 (accidental strangulation and suffocation) - to define inclusion since cause of death is publicly available death certificate data. Use of death scene investigation or other data from the Child Death Review team was not permitted under the ethics approval for this study. The non-fatal unsafe sleep events were defined as cases in which unsafe sleep conditions resulted in evaluation at UPMC Children's Hospital of Pittsburgh with a referral being made to Child Protective Service (CPS). Data about the circumstances of the event were collected from the electronic health record. The outcome of the CPS reports was not known. Cases which met these criteria from 7/1/17–12/31/22 were included if a PRM score was able to be calculated.

2.1.2. Controls

The control group includes all Allegheny County births from 9/1/2020–8/31/22.

2.1.3. Hello baby PRM construction and validation

The methodology for the construction and validation including the list of the predictors, the area under the curve with respect to predicting foster care placement and the sensitivity and specificity of the Hello Baby PRM at various thresholds has been described previously and is available on the Allegheny County website (Vaithianathan et al., 2019).

Hundreds of variables were used to the train the PRM using births in Allegheny County between 2012 and 2015 ($N = 52{,}520$). Potentially predictive features were coded for each birth record. In addition, features available from the birth certificate (e.g., birthweight, maternal age, maternal smoking status), the use of homeless services, and court, jail and CPS system contacts for mother and father (if identified on the birth certificate) were coded. A Lasso model was trained to predict removal from the home by CPS within the first 3 years of life. Removals were used as the predicted outcome because the Hello Baby PRM tool was focused on preventing long-term CPS involvement. The variables in the final model are listed in Appendix A. Since many of the predictors are interrelated and 'bundled together' and there can be missing data, there is no list of predictors that map to a high PRM score. The methodology for bundling predictors are described in the technical paper about the PRM validation (Vaithianathan et al., 2017). Multiple attributes, complicated weights and then a transformation of the weighted predictors is used to develop a PRM score. This complexity is a disadvantage insofar as it is not possible to provide simple causal explanations for a given PRM score for an individual child, but it is essential to get the model's high predictive accuracy and generate scores that allow universal ranking of children according to estimated risk of removal. The PRM validation also incorporated and allowed for subjects with missing data. There is, however, a minimum requirement for a score to be generated; if there is no birth record, no PRM score will be calculated.

2.2. Data collection

Data collected from Allegheny County's integrated Data Warehouse was examined for cases and controls. The source of data for both cases and controls was the same: birth certificate records, child welfare data, and death certificates. The following data were collected: race, gender, insurance at birth (public vs. private), maternal age, gestational age, maternal and paternal education level, and presence or absence of maternal smoking during pregnancy. For the cases, age at event was also collected. Race is included as a variable because of the recognized and persistent relationship between race and the rate of SUID (Boyer et al., 2022; Singh & Yu, 2019; Yamada et al., 2021). It is used as a social construct in the context of our work and is not a predictor in the Hello Baby PRM model.

A Hello Baby PRM score at the time of birth was calculated for all cases and controls. Prior to 9/1/20, the score was calculated retroactively. Beginning 9/1/20, the score was automatically assigned at birth and used to proactively offer a range of intensive services to families designed to address the modifiable risk factors for foster care placement which are identified in individual families.

2.2.1. Statistical analysis

Descriptive statistics were used to describe the populations. Chi-square, *t*-test, and Mann Whitney *U* test were used to compare populations. A $p < 0.05$ was considered significant. To address the possible concern about different variances between cases and controls, a Fligner–Policello test was used in addition to the Mann-Whitney-U test to compare the SUID/unsafe sleep group to controls. An area under the curve was calculated to evaluate the predictive value of a PRM score of 20 for SUID/unsafe sleep events. The study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines. This evaluation was approved by Auckland University of Technology Ethics Committee.

3. Results

Between 7/1/17 and 12/31/22, 101 infants died from SUID ($n = 64$) or had an unsafe sleep event ($n = 37$). In all 37 unsafe sleep events, the infant was on a bed, couch, futon, or air mattress. All infants had at least one other unsafe sleep condition including being in a parent's arm or on a parent's chest ($n = 7$), in bed with another person(s) ($n = 23$), and/or prone ($n = 11$). Among the 30 cases in which the infant was sleeping with another person, the other person included a sibling in 27 % (8/30) of cases.

Of the 37 unsafe sleep events, 32 resulted in injuries. Injuries included skull fracture with or without underlying extra-axial hemorrhage (n = 7), non-skull fracture ($n = 8$), unresponsiveness with apnea requiring resuscitation (n = 2), or dermatologic injuries including bruising and subconjunctival hemorrhages ($n = 15$). The five unsafe sleep events without injuries raised concern about safety and resulted in a report to CPS; in two cases, a parent was found co-sleeping by a home visitor, in one case, the ED visit was the second in two weeks for a fall during co-sleeping and in two cases, no details were available which would explain why there was a referral to CPS even though there were no injuries.

Of the 101 cases, 99 were able to be given a PRM score; of these, 62 were SUID and 37 were infants who had an unsafe sleep event. The other two infants were born outside of Allegheny County so a birth certificate was unavailable and, therefore, a PRM score could not be calculated.

Children with non-fatal unsafe sleep events were similar to children with SUID for all demographic variables evaluated except for two: they were older than infants with SUID and were more likely to be first children ($p < 0.00$ for each) (Table 1). They also had similar PRM scores (Fig. 1). Among children with SUID or an unsafe sleep event, 50 % had a PRM score of 17–20. This is markedly higher than for the control group of 23,633 births, where 16.7 % of births had a score 17–20 ($p < 0.00$). In addition, infants with a score

Table 1
Demographic, social factors, and PRM risk score among infants with SUID, non-fatal unsafe sleep event and controls.

	SUID ($N = 62$)	Non-fatal unsafe sleep event (n = 37)	*p*-value (SUID vs. unsafe sleep)	Control (n = 23,633)	p-value (cases vs. controls)
Median age in months (range)	2.6 (0.4–9.8)	5.1 (0.4–10.5)	**< 0.00**	N/A	N/A
Median Hello Baby PRM risk score	18	16	NS	10	**<0.01**
% Male	52 %	51 %	NS	51 %	NS
% White	45 %	35 %	NS	37 %	NS
% Public insurance	74 %	62 %	NS	26 %	**<0.01**
% full-term (≥37 weeks gestation)	74 %	86 %	NS	91 %	NS
% only/first children	21 %	51 %	**0.03**	43 %	NS
% maternal smoking (during pregnancy)	44 %	35 %	NS	8 %	**<0.01**
Mean (SD) maternal age	28.4 (6.1)	27.3 (6.0)	NS	30.9	NS
% mothers married	32 %	30 %	NS	67.0 %	**<0.01**
% of mothers with college education	19 %	30 %	NS	65.3 %	**<0.01**
% of fathers with college education	15 %	22 %	NS	56 %	**<0.01**

Note: Missing data was ≤10 % for all variables except first children (11 % for controls), maternal smoking (19 % for controls) and father's education (30 % for cases and 45 % for controls).
Bold signifies p < 0.05.

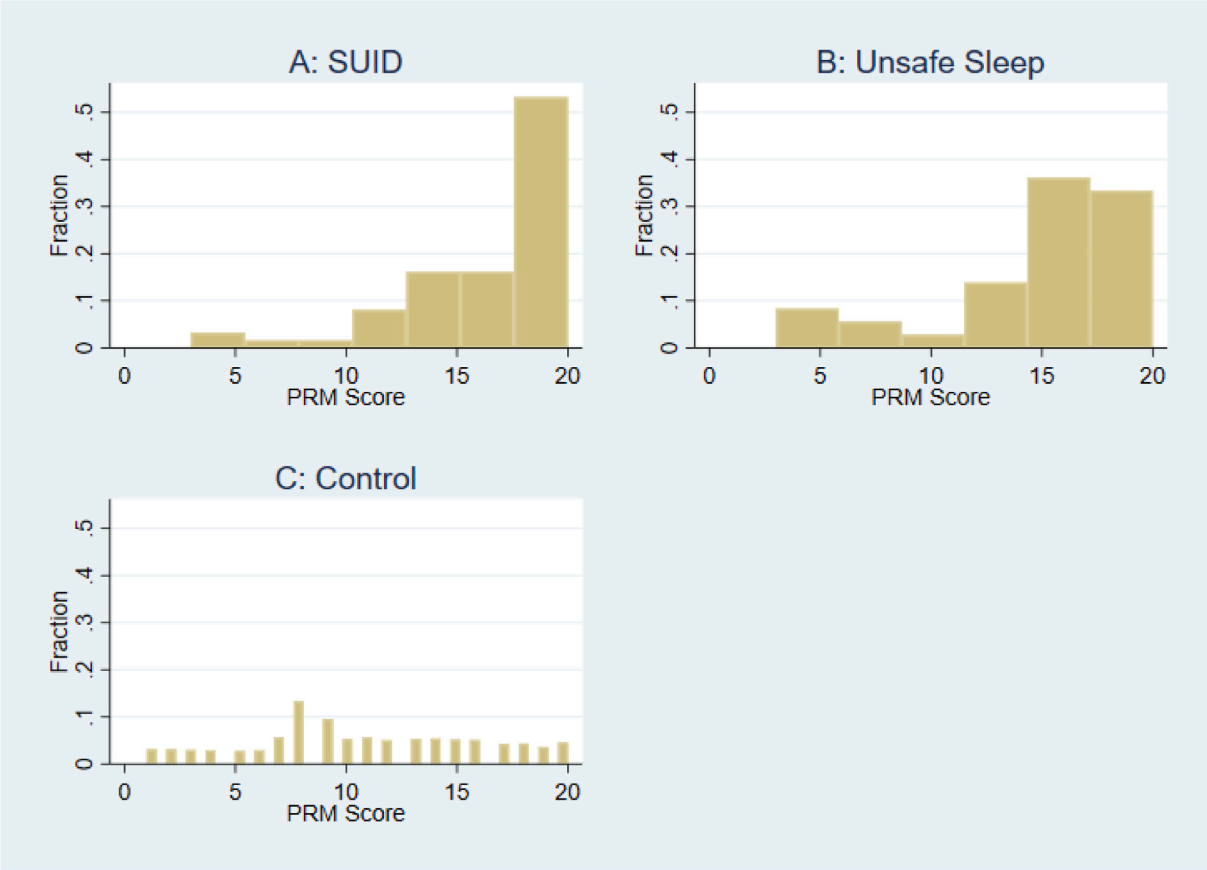

Fig. 1. Histogram of PRM scores for (a) Infants with SUID ($n = 62$) (b) Infants with a non-fatal unsafe sleep events ($n = 37$) and (c) a control cohort of county-wide births ($n = 23{,}633$).
Abbreviations: SUID; sudden unexplained infant death.

of 20 accounted for 31 % of all the SUID/unsafe sleep events but only 5 % of the birth cohort. ($p < 0.00$) (Fig. 1). For the 2020–2021 birth cohort, the rate of SUID/unsafe sleep event was 1.1 per 1000. The risk of SUID/unsafe sleep event in an infant with a score of 20 in this cohort was 7.8 per 1000. The area under the curve for prediction of SUID/unsafe sleep event in infants with a score of 20 was 0.7865 (95 % CI: 0.690–0.883).

The birth cohort (control group) was significantly different from the case group in multiple respects: controls had lower rates of public insurance and maternal smoking during pregnancy and higher rates of married mothers and mothers and fathers with a college education ($p < 0.00$ for each, Mann-Whitney and Fligner–Policello) (Table 1). They also had significantly lower PRM scores ($p < 0.00$) (Fig. 1).

4. Discussion

This is the first study, to our knowledge, to demonstrate that a PRM which is currently in use for preventing child abuse and neglect can also identify children at increased risk of SUID and non-fatal unsafe sleep events. The Hello Baby PRM identifies which families are at highest risk for the outcome of interest which prior to this publication, had been placement in foster care. We have now demonstrated that the same PRM also predicts a different outcome: SUID or unsafe sleep events. The approach of re-purposing a PRM developed for one use for another use is something our group has done previously (Vaithianathan, Putnam-Hornstein, et al., 2020). In our 2020 publication in *JAMA Pediatrics*, we demonstrated that the Allegheny Family Screening Tool - which was trained on the outcome of foster care placement - was correlated with another outcome, hospital admission.

Once the PRM identifies the highest risk families, the next step is to evaluate for modifiable/mutable risk factors in individual families and then attempt to address them. The evaluation of the Hello Baby intervention - addressing the risk factors identified in high-risk families - is ongoing with the goal of understanding if the combination of a PRM and intervention is effective in improving the primary outcome of interest: placement in foster care by 3 years of age. Given the finding in the current study - that the Hello Baby PRM can also identify infants at the time of birth who are high-risk of SUID and unsafe sleep events - evaluation and intervention for the modifiable risk factors for SUID and unsafe sleep event would be the next step.

Prediction - which is what the PRM does - is about finding a population where there is sufficiently high prevalence of the outcome of interest, entry into foster care, SUID or unsafe sleep events - that it is cost-effective to assess for modifiable risk factors on an individual basis. Targeting of high-risk groups for intensive interventions is important for all programs since there are limited resources and not every program can be universal. Programs such as Nurse Family Partnership (NFP) or Early Intervention or WIC, for example, are not offered to everyone, but only to a subset of children and families who have been identified as having the most potential to benefit from these programs.

The lower proportion of infants with SUID who are first born children is interesting and consistent both with the concept that parents have limited capacity to attend to demands and with data which demonstrate higher infant injury rates and mortality rates among non-first-born children (*Linked birth/infant death records on CDC WONDER online database*, n.d.; Ahrens et al., 2017; Bijur et al., 1988; Lawson & Mace, 2009; Taylor et al., 2007). One of the reasons proposed for these findings is divided parental attention. Similarly, when there is only one child in the home, parent(s) may be more able to follow safe sleep guidelines, but when there are multiple children, parents are more sleep-deprived and may be more likely to bring an infant into bed with them so they can get more sleep. It may also suggest that when families do not practice safe sleep with a first child and that child does not have a bad outcome, they are more likely to continue with these sleep practices for future children; the more 'exposures' to unsafe sleep, the more likely there is to be a bad outcome (Herman et al., 2015). Finally, a misperception that co-sleeping with siblings (vs. co-sleeping with an adult) is safe might also account for the increase in non-first born SUID.

Given the similarities in the PRM scores for child maltreatment and SUID/unsafe sleep events, one might hypothesize that some of the approaches to decreasing child maltreatment which may impact multiple social determinants of health may also influence SUID and unsafe sleep events. Increasing income through the offering of cash benefits, for example, has been extensively evaluated as an approach to decreasing child maltreatment (Bruckner & Catalano, 2006; Maxfield & Thomson, 2023). Cash benefits may decrease financial stress allowing mothers to delay returning to the workforce and may also allow them to be able to continue to act on safe sleep messaging. As discussed previously as it relates to first vs. later born infants, parents have limited capacity to attend to demands on them; remembering to put an infant to sleep in a crib on their back may be overlooked among caregivers distressed by economic stressors in the same way that it may be overlooked in a house with multiple young children.

This is the first study, to our knowledge, which has included a non-fatal unsafe sleep injury group. Among the measurable variables, the SUID and unsafe sleep event groups are similar except in age and birth order. There, of course, may be unobservable and/or non-measured factors which differ between the groups but the similarities between the PRM scores in the two groups suggests that there are many similarities. While the lack of a statistical difference in the PRM scores between children with a SUID and an unsafe sleep event could be due to a type II error, both the SUID group and the unsafe sleep event group have higher PRM scores than controls which are critical for identification of their risk status. The similarities between the groups for the measured variables led us to decide to combine them into a single group for analysis. We provided data about the SUID and unsafe sleep events separately, however, so that the reader can see all the data. Larger sample sizes and additional data from other communities would help to understand differences between infants with SUID and unsafe sleep events.

This study has several strengths. The fact that the Hello Baby PRM has already been deployed and uses data which are available and being used for one reason - prediction of home removal due to child maltreatment - to address another cause of infant morbidity and mortality - SUID and unsafe sleep events - means that this approach to identifying infants at high risk of SUID and unsafe sleep events could be implemented immediately. While much of the literature is focused on building hypothetical models, this study is unique in that it evaluates a previously deployed PRM. The fact that the Hello Baby PRM score is automated and does not require clinician input is also a strength as it relates to rapid implementation. The study's relatively large sample size for a relatively rare event is also a strength as is the comparison to a local control group.

There are a number of limitations. The lack of generalizability is one; many counties and jurisdictions do not have the ability to calculate a PRM score automatically with existing data. However, many CPS agencies around the country have already integrated data to improve decision-making by front line workers (Auspos, 2017; Improving Child and Family Services Through Integrated Data Systems, 2017). Development of a PRM is the next step in using integrated data to improve child outcomes. Bringing an algorithm such as Hello Baby to clinical use is a long process which requires community participation and strict ethical considerations (*Report of the Pittsburgh Task Force on Public Algorithms*, 2023; Vaithianathan et al., 2021). The Hello Baby PRM has undergone ethical review both by an independent firm (Daro, 2019; Veale, 2019) and by the Task Force on Public Algorithms in Allegheny County (*Report of the Pittsburgh Task Force on Public Algorithms*, 2023). External validation of similar PRMs in other jurisdictions is an important next step.

Another limitation was the control group includes a 2-year birth cohort rather than a cohort which covers the entire time period; characteristics of the birth cohort such as maternal age and rate of maternal smoking are very stable over a 5-year or even 10-year study period and thus the smaller birth cohort is unlikely to affect the results or conclusions. The difficulty translating risk into action is always a limitation of this type of work; while the ability to identify the highest risk infants for post-neonatal mortality represents an opportunity for intervention, the families at highest risk are often the most difficult to engage with services designed to decrease family specific modifiable risks. The amount of missing data on the birth certificates may also be a limitation since it could influence the comparison between groups. Because birth-certificate data are self-reported, missing data particularly for variables such as maternal smoking may be non-random. As a result, differences between cases and controls may be smaller or larger than reported.

A final limitation relates to the subgroup of cases with non-fatal unsafe sleep events which are unlikely to be representative of all infants with non-fatal unsafe sleep events. There are certainly additional infants with non-fatal unsafe sleep events who did not meet the inclusion criteria – those that didn't come to the hospital after an unsafe sleep event and those who came to the hospital but were not referred to CPS. Infants who came to the hospital but were not referred to CPS would be part of the control group but if they had a high PRM score, they would be included in the high-risk group and would be targeted for any interventions designed to decrease the outcome of interest.

5. Conclusion

Our data demonstrate – perhaps not surprisingly – that infants who are at risk of being placed in foster care are also at risk of a range of other adverse outcomes, including SUID and non-fatal unsafe sleep events. The Hello Baby PRM tool may, therefore, be a potentially useful approach to identify infants who are at heightened risk of SUID and non-fatal unsafe sleep events and who can be targeted for prevention programs which address family-specific modifiable risk factors.

Unlike more time-consuming approaches to identifying high-risk infants at birth, the PRM is automated, it requires no manual data entry by a clinician and provides a way of stratifying the whole population at the time of birth which allows for targeted and tailored interventions.

CRediT authorship contribution statement

Julia Reuben: Writing – review & editing, Resources, Methodology, Formal analysis, Data curation. **Rhema Vaithianathan:** Writing – review & editing, Software, Resources, Methodology, Formal analysis. **Rachel Berger:** Writing – review & editing, Writing – original draft, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

Data availability

The data that has been used is confidential.

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The study data have not been presented previously.

Appendix A. Supplementary data

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